

Today: Eigenvalues and Eigenvectors

Analytic Geometry

Tomorrow: Quiz 8 on eigenvalues and eigenvectors
Review for final exam

9.3 Example: so. $A = \begin{bmatrix} 0 & 0 \\ 1 & -3 \end{bmatrix}$ triangular $\Rightarrow \lambda = 0, -3$

$\lambda = 0:$

$$\begin{bmatrix} 0 & 0 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$u_1 - 3u_2 = 0$$

$$1u_1 = 3u_2$$

$$\vec{u} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

$\lambda = -3$

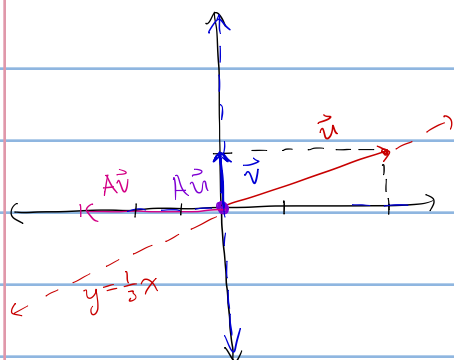
$$\begin{bmatrix} 0+3 & 0 \\ 1 & -3+3 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 0 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$3v_1 + 0v_2 = 0$$

$$1v_1 = 0v_2$$

$$\vec{v} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$



$$A\vec{u} = \lambda\vec{u}$$

$$\begin{bmatrix} 0 & 0 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$y = \frac{1}{3}x$$

$$x = 0$$

$$A\vec{v} = \lambda\vec{v}$$

$$\begin{bmatrix} 0 & 0 \\ 1 & -3 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -3 \end{bmatrix}$$

$$54.) A = \begin{bmatrix} 3 & 6 \\ -1 & -4 \end{bmatrix} \quad \tau = 3-4 = -1$$

$$\Delta = (3)(-4) - (-1)(6) = -12 + 6 = -6$$

$$\lambda^2 - \tau\lambda + \Delta = 0 \quad \begin{vmatrix} 3-\lambda & 6 \\ -1 & -4-\lambda \end{vmatrix} = (3-\lambda)(-4-\lambda) - 6(-1) = 0$$

$$\lambda^2 + \lambda - 6 = 0 \quad \begin{vmatrix} -1 & -4-\lambda \end{vmatrix}$$

$$(\lambda-2)(\lambda+3) = 0 \quad = -12 + 4\lambda - 3\lambda + \lambda^2 + 6 = 0$$

$$\lambda = 2, -3 \quad = \lambda^2 + \lambda - 6 = 0$$

$$(\lambda-2)(\lambda+3) = 0$$

$$\lambda = 2, -3$$

$$\lambda = 2:$$

$$\begin{bmatrix} 3-2 & 6 \\ -1 & -4-2 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 6 \\ -1 & -6 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$u_1 + 6u_2 = 0$$

$$u_1 = -6u_2$$

$$\vec{u} = \begin{bmatrix} -6 \\ 1 \end{bmatrix} \quad \vec{u} = \begin{bmatrix} 6 \\ -1 \end{bmatrix}$$

$$A\vec{x} = \lambda\vec{x}$$

$$\lambda = -3:$$

$$\begin{bmatrix} 3-(-3) & 6 \\ -1 & -4-(-3) \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 6 & 6 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$6v_1 + 6v_2 = 0$$

$$v_1 = -v_2$$

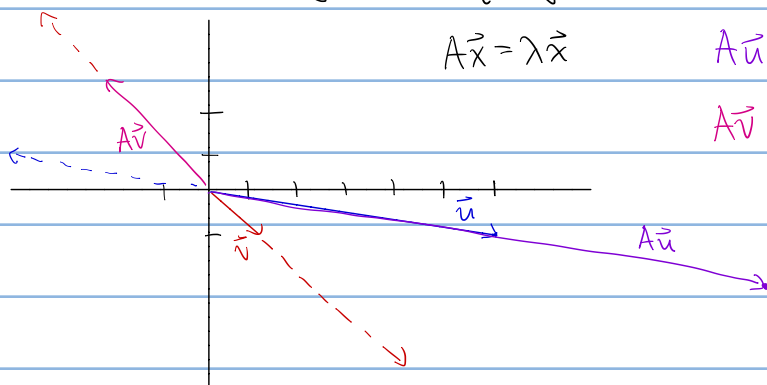
$$\vec{v} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$A\vec{u} = 2\vec{u}$$

$$A\vec{v} = -3\vec{v}$$

$$y = -\frac{1}{6}x$$

$$y = -x$$

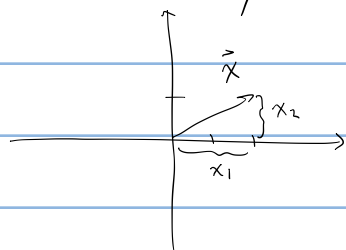


$$\begin{bmatrix} 3 & 6 \\ -1 & -4 \end{bmatrix} \begin{bmatrix} 6 \\ -1 \end{bmatrix} = \begin{bmatrix} 3 \cdot 6 + 6 \cdot (-1) = 12 \\ 6 \cdot (-1) + (-4) \cdot (-1) = -2 \end{bmatrix}$$

$$= 2 \begin{bmatrix} 6 \\ -1 \end{bmatrix}$$

9.4 Analytic Geometry

$$\vec{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

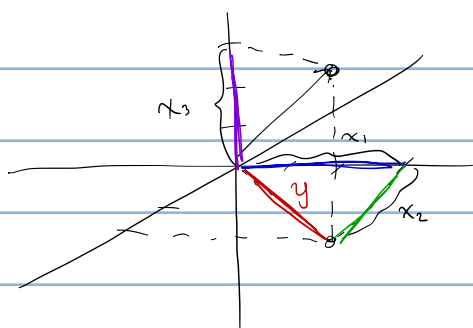


$|\vec{x}|$ - length or magnitude of vector

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$|\vec{x}| = \sqrt{x_1^2 + x_2^2}$$

In 3 dimensions



$$\vec{x} = \begin{bmatrix} 3 \\ 2 \\ 3 \end{bmatrix}$$

$$|\vec{x}| = \sqrt{3^2 + 2^2 + 3^2} \\ = \sqrt{22}$$

$$y = \sqrt{x_1^2 + x_2^2} \\ |\vec{x}| = \sqrt{y^2 + x_3^2} \\ = \sqrt{(\sqrt{x_1^2 + x_2^2})^2 + x_3^2} \\ = \sqrt{x_1^2 + x_2^2 + x_3^2}$$

In n-dimensions

$$\vec{x} \in \mathbb{R}^n \quad |\vec{x}| = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2}$$

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

$$= \left(\sum_{i=1}^n x_i^2 \right)^{1/2}$$

Unit vector of $\vec{x}$

$\hat{x}$ - unit vector in the direction of $\vec{x}$

$$\hat{x} = \frac{\vec{x}}{|\vec{x}|} \quad \vec{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \quad |\vec{x}| = \sqrt{2^2 + 1^2} = \sqrt{5}$$
$$\frac{\vec{x}}{|\vec{x}|} = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$|\hat{x}| = \sqrt{\left(\frac{2}{\sqrt{5}}\right)^2 + \left(\frac{1}{\sqrt{5}}\right)^2} = \sqrt{\frac{4}{5} + \frac{1}{5}} = \sqrt{1} = 1$$

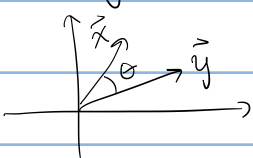
$$\vec{x} = \begin{bmatrix} 3 \\ 2 \\ 3 \end{bmatrix} \quad |\vec{x}| = \sqrt{22} \quad \sqrt{\left(\frac{3}{\sqrt{22}}\right)^2 + \left(\frac{2}{\sqrt{22}}\right)^2 + \left(\frac{3}{\sqrt{22}}\right)^2}$$
$$\sqrt{\frac{9}{22} + \frac{2}{22} + \frac{9}{22}} = \sqrt{1}$$
$$= 1$$

The Dot Product (Scalar Product)

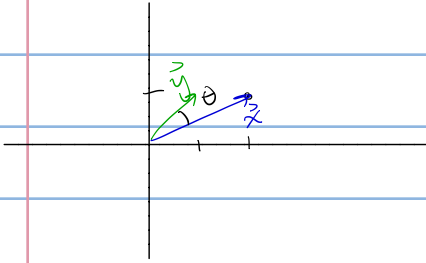
$$\vec{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad \vec{y} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \quad \vec{x} \cdot \vec{y} = 1 \cdot 3 + 2 \cdot 2 = 3 + 4 = 7$$

$$\vec{x} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \quad \vec{y} = \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix} \quad \vec{x} \cdot \vec{y} = 1 \cdot 3 + 2 \cdot 1 + 1 \cdot 4 = 3 + 2 + 4 = 9$$

$$\vec{x} \cdot \vec{y} = \vec{x}^T \vec{y} \quad \begin{bmatrix} 1 & 2 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \\ 4 \end{bmatrix} = 3 \cdot 1 + 2 \cdot 1 + 4 \cdot 1 = 9$$

$$\vec{x} \cdot \vec{y} = |\vec{x}| |\vec{y}| \cos \theta \Rightarrow \cos \theta = \frac{\vec{x} \cdot \vec{y}}{|\vec{x}| |\vec{y}|}$$
$$\theta = \cos^{-1} \left(\frac{\vec{x} \cdot \vec{y}}{|\vec{x}| |\vec{y}|} \right)$$


$$\vec{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \quad \vec{y} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$



$$\vec{x} \cdot \vec{y} = 2 \cdot 1 + 1 \cdot 1 = 3$$

$$|\vec{x}| = \sqrt{2^2 + 1^2} = \sqrt{5}$$

$$|\vec{y}| = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\theta = \cos^{-1}\left(\frac{3}{\sqrt{5} \cdot \sqrt{2}}\right) = \cos^{-1}\left(\frac{3}{\sqrt{10}}\right) \approx 18.4^\circ$$